

Practical 1

About this unit

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Practice Lab Assignment

Unit • 100% completed



Lab Assignment

Unit • 100% completed



1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

m is the mass of the object (in kilograms).

v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1
2 mass = float(input())
3 velocity = float(input())
4
5 momentum = mass * velocity
```

Average time

0.003 s

2.50 ms

Maximum time

0.004 s

4.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

5.0

10.0

50.00kgm/s

Actual output

5.0

10.0

50.00kgm/s

Test case 2

1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2
3 n = int(input())
4
5 if 1 <= n <= 9:
6     print(n ** 2)
7
8 elif 10 <= n <= 99:
9     print(f"{n ** (1/2):.2f}")
10
11 elif 100 <= n <= 999:
12     print(f"{n ** (1/3):.2f}")
13
14 else:
15     print("Invalid")
```

1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = int(input())
2
3 rev = 0
4
5 while n > 0:
6     digit = n % 10
7     rev = rev * 10 + digit
8     n = n // 10
9
10 print(rev)
```

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`
- The second line contains the second date in the format `YYYY-MM-DD`

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2
3 # Input dates
4 date1 = input().strip()
5 date2 = input().strip()
6
7 # Convert string to datetime objects
8 d1 = datetime.strptime(date1, "%Y-%m-%d")
9 d2 = datetime.strptime(date2, "%Y-%m-%d")
10
11 # Calculate difference
12 difference = (d2 - d1).days
13
14 # Output result
15 print(difference)
```

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 n = int(input())
2
3 for i in range(1, 11):
4     print(f"{n} x {i} = {n*i}")
5
6
7
```

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage.

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

```
1 n = int(input())
2 marks = list(map(int, input().split()))
3
4 fail = False # make sure this is OUTSIDE the loop
5
6 for m in marks:
7     if m < 40:
8         fail = True
9         break
10
11 if fail:
12     print("Fail")
13 else:
14     avg = sum(marks) / n
15     print(f"Aggregate Percentage: {avg:.2f}")
16
17     if avg > 75:
18         print("Grade: Distinction")
19     elif avg >= 60:
20         print("Grade: First Division")
21     elif avg >= 50:
22         print("Grade: Second Division")
23     elif avg >= 40:
24         print("Grade: Third Division")
```

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2
3     if n == 1:
4         return 0
5     elif n == 2:
6         return 1
7     else:
8         return fibonacci(n-1) + fibonacci(n-2)
9
10
11 n = int(input())
12 for i in range(1, n+1):
13     print(fibonacci(i), end=" ")
14
```

Average time

0.007 s

6.75 ms

Maximum time

0.018 s

18.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

0 1 1 2 3

Actual output

0 1 1 2 3

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n = int(input())
2
3 for i in range(1, n + 1):
4     for j in range(1, i + 1):
5         print(j, end=" ")
6     print()
7
8
```

Average time0.003 s2.50 ms

Maximum time0.003 s3.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output